

Effectiveness and Return On Investment (ROI) on Aerodynamic Technologies for Heavy Duty Vehicles

The transportation sector contributes about 28% of U.S. greenhouse gas (GHG) emissions (US EPA, 2022) [1]. While passenger and light-duty vehicles have received extensive attention in emission reduction efforts, heavy-duty vehicles have historically been less emphasized. Recently, programs such as the Environmental Protection Agency's (EPA) voluntary SmartWay program and the California Air Resources Board's (CARB) tractor-trailer GHG regulation have provided incentives to improve the environmental performance of heavy-duty trucks [2].

The SmartWay program develops and validates fuel-saving technologies while establishing benchmarks for granting "SmartWay" designation. CARB has built on this framework to implement regulations that increase the efficiency of heavy-duty vehicles in California. Key technologies promoted include aerodynamic devices (skirts, under trays, boat tails, gap reducers) and low rolling-resistance tires, all of which reduce drag and fuel consumption, thereby lowering GHG emissions and transportation costs.

To determine the effectiveness and ROI of these technologies, an estimate of the fuel savings, cost savings and payback period is shown below.

Assumptions [2,3]:

Annual miles: 35,000 miles; Fuel consumption: 6.5mpg; annual fuel use: 5,385 gallons; diesel price: \$3.75; fuel saving with skirt: 7% (best case); cost of installation: \$1,100.

Cost of use of fuel (annual) = $\$3.75 \times 5,385 = \$20,193.8/\text{yr}$

Percentage fuel savings = $0.07 \times 5385 = 376.95$ gallons

Percentage cost savings = $\$3.75 \times 376.95 = \1414

Payback period = $\frac{\text{Investment cost}}{\text{net annual savings}} = \frac{\$1,100}{\$1,414} = 0.779 \approx 0.8 \text{ years} \approx 10 \text{ months}$

Aerodynamic technologies usually have a life span of about 5 years, net saving over 5 years will be

$$\$1,414 \times 5 - \$1,100 = \$5,970$$

The estimation above demonstrates that aerodynamic technologies, with a payback period of about 10 months, deliver a high return on investment and significantly improve fuel efficiency compared to baseline long-haul trailers without such devices. Their effectiveness is reflected not only in cost savings but also in widespread adoption: approximately 40% of trailers sold in the U.S. are equipped with skirts, and about one-quarter feature one or more aerodynamic devices.

References:

1. US EPA (2022). Inventory of U.S. Greenhouse Gas Emissions and Sinks. <https://www.epa.gov/ghgemissions/inventory-us-greenhouse-gas-emissions-and-sinks> Accessed on September 28, 2025.
2. Sharpe, B., & Roeth, M. (2014). Costs and adoption rates of fuel-saving technologies for trailers in the North American on-road freight sector. *The International Council on Clean Transportation*, www.theicct.org.
3. NACFE (2018). Trucking efficiency confidence report: trailer aerodynamics executive summary. https://nacfe.org/wp-content/uploads/2018/02/RMICWR_Trucking_ExecSummary_Aero_FINAL.pdf Accessed on September 28, 2025